



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 20: Welfare Analysis

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 19

Last lecture, we studied **General Equilibrium models**

- General Equilibrium with Two Goods → Edgeworth Box
- Remember the Production Possibility Frontier (PPF)
- Did a comparative statics analysis
- Derived a mathematical model for general equilibrium prices
- Learnt Walras' proof of existence of general equilibrium prices

In today's lecture, we will

- Start Welfare Economics and Economic Efficiency

Announcements

- Remember Cheat Sheet for Test 3
- Face-to-face class assistantship today at B22

Determination of Equilibrium Prices

Competitive Markets

PPF Example

Following Activity 3 (lecture 18), where the production of x and y depends only on labour input (total labour supply is 100) with:

$$x = f(l_x) = l_x^{0.5}$$

$$y = f(l_y) = l_y^{0.5}$$

Thus, the PPF was given by: $x^2 + y^2 = 100$, for $x, y \geq 0$ and $RPT = x/y$

We will assume that the typical individual's utility function is given by

$$U(x, y) = x^{0.5} y^{0.5}$$

- a) Calculate the demand functions for the two goods
- b) Find the base-case equilibrium (x^* , y^* and a relationship between P_x and P_y)
- c) Can you find an expression for the individual's budget constraint? (or his/her total income)
- d) If there is a technical improvement in x production so that the production function is $x = 2l_x^{0.5}$. Determine the new PPF, RPT and equilibrium.
- e) If consumer preferences switch to favour good y as $U(x, y) = x^{0.1} y^{0.9}$, determine the new demand functions and equilibrium with the old PPF.

PPF Example: Solution

- a) Calculate the demand functions for the two goods

In order to find the demand functions, we have to

$$\text{Max } U(x, y) = x^{0.5}y^{0.5}$$

$$\text{Subject to: } P_x x + P_y y = I$$

$$\text{Since it is an interior solution: } \frac{MU_x}{MU_y} = \frac{P_x}{P_y} \rightarrow \frac{y}{x} = \frac{P_x}{P_y}$$

Replacing this condition in the budget constraint:

$$P_x x + P_y y = I$$

$$P_x x + P_x x = I$$

$$x_m = \frac{I}{2P_x}, y_m = \frac{I}{2P_y}$$

PPF Example: Solution

- b) Find the base-case equilibrium (x^* , y^* and a relationship between P_x and P_y)

Profit maximization by firms require:

$$\frac{P_x}{P_y} = \frac{MC_x}{MC_y} = RPT = \frac{x}{y}$$

Utility-maximizing demand requires:

$$\frac{P_x}{P_y} = MRS = \frac{MU_x}{MU_y} = \frac{y}{x}$$

Thus, equilibrium requires: $\frac{y}{x} = \frac{x}{y} \rightarrow x = y$

Replacing in the PPF: $x^2 + y^2 = 100$

$$x^* = y^* = \sqrt{50}, \quad \frac{P_x}{P_y} = 1$$

PPF Example: Solution

- c) Can you find an expression for the individual's budget constraint? (or his/her total income)

The budget constraint faced by individuals is not especially transparent. We will consider all prices in terms of the wage rate (w)

Because general equilibrium models must obey the national income identity, we assume that consumers are also shareholders in the two firms and treat these profits as part of their spendable income.

Hence, total consumers income is:

$$\text{Total Income} = \text{Labour Income} + \text{Profits}$$

PPF Example: Solution

- c) Can you find an expression for the individual's budget constraint? (or his/her total income)

Since the total labour supply is 100 $\rightarrow$ **total labour income is $100w$**

Because of the diminishing returns assumed for production, each firm also earns profits.

For Firm x, the total cost function is:

$$C(w, x) = wl_x = wx^2$$
$$P_x = MC_x = 2wx = 2w\sqrt{50}$$

Profits for firm x are, therefore:

$$\pi_x = (P_x - AC_x) \cdot x = (2wx - wx) \cdot x = wx^2 = 50w$$

Similarly, for firm y: $\pi_y = 50w$

PPF Example: Solution

- c) Can you find an expression for the individual's budget constraint? (or his/her total income)

Total Income = Labour Income + Profits

$$\text{Total Income} = 100w + 2(50w) = 200w$$

This income will just permit consumers to spend $100w$ on each good by buying $\sqrt{50}$ units at a Price of $2w\sqrt{50}$, so the model is internally consistent.

PPF Example: Solution

- d) If there is a technical improvement in x production so that the production function is $x = 2l_x^{0.5}$.
Determine the new PPF, RPT and equilibrium

This technical improvement is a SHIFT IN SUPPLY

We know that $l_x + l_y = 100 \rightarrow \frac{x^2}{4} + y^2 = 100 \rightarrow$ new PPF

$$\frac{x^2}{4} + y^2 = 100 \quad /d()$$

$$\frac{2x \cdot dx}{4} + 2y \cdot dy = 0$$

$$RPT = -\frac{dy}{dx} = \frac{x}{4y}$$

PPF Example: Solution

- d) If there is a technical improvement in x production so that the production function is $x = 2l_x^{0.5}$.
Determine the new PPF, RPT and equilibrium

New equilibrium:

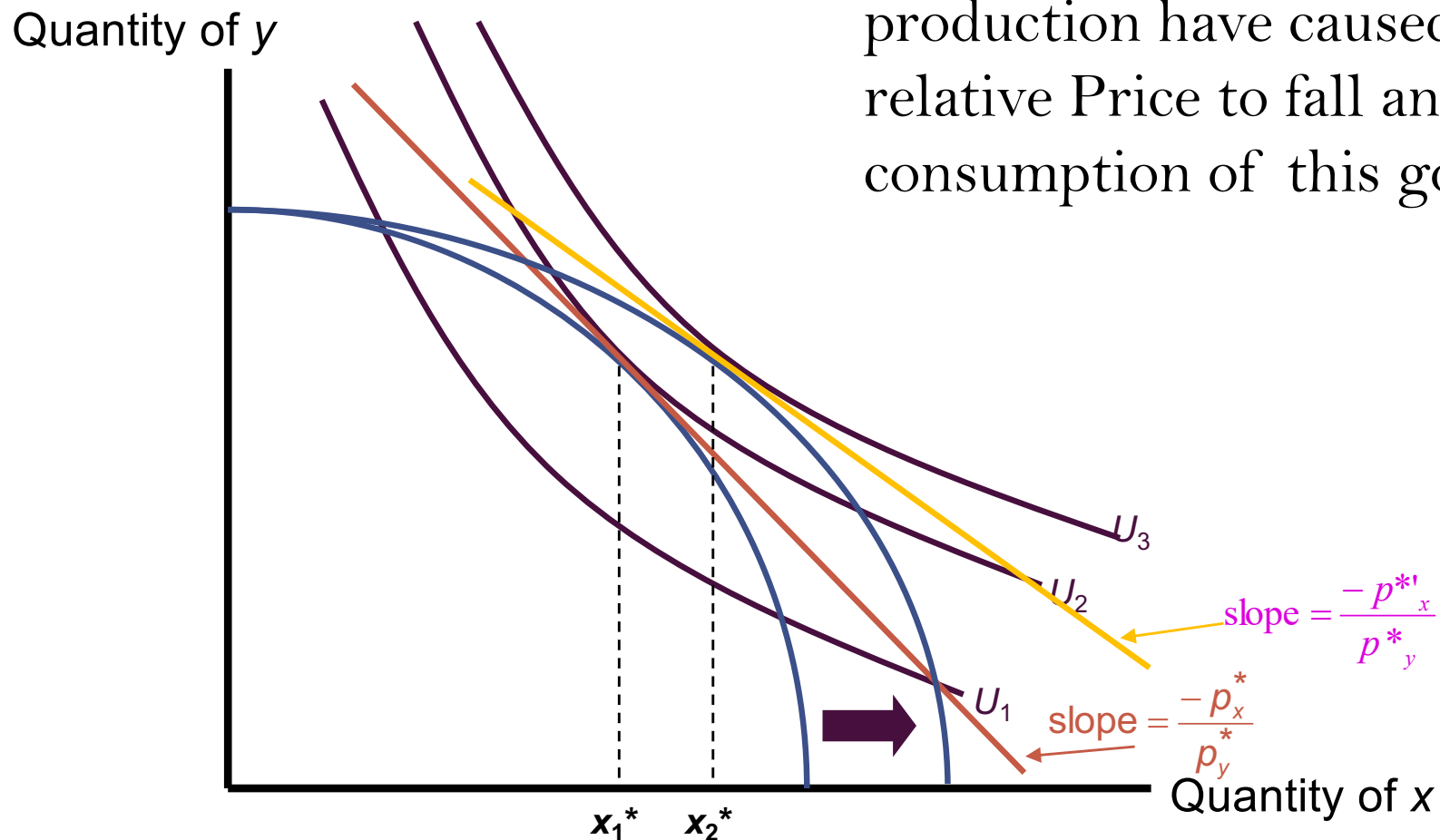
Supply: $\frac{P_x}{P_y} = \frac{x}{4y}$ Demand: $\frac{P_x}{P_y} = \frac{y}{x}$

Thus, $x^2 = 4y^2 \rightarrow$ Replacing in PPF

$$\frac{x^2}{4} + y^2 = 100$$

New equilibrium: $x^* = 2\sqrt{50}$, $y^* = \sqrt{50}$, $\frac{P_x}{P_y} = \frac{1}{2}$

PPF Example: Solution



PPF Example: Solution

- d) If there is a technical improvement in x production so that the production function is $x = 2l_x^{0.5}$. Determine the new PPF, RPT and equilibrium

Technical improvements make consumers better off:

$$U_1 = x^{0.5}y^{0.5} = \sqrt{50} = 7.07$$

$$U_2 = x^{0.5}y^{0.5} = (2\sqrt{50})^{0.5}(\sqrt{50})^{0.5} = \sqrt{2}\sqrt{50} = 10$$

PPF Example: Solution

- e) If consumer preferences switch to favor good y as $U(x, y) = x^{0.1}y^{0.9}$, determine the new demand functions and equilibrium with the old PPF

If consumers were to switch to favor good y as:

$$U(x, y) = x^{0.1}y^{0.9}$$

New demand functions:

$$x = \frac{I}{10P_x}, y = \frac{9I}{10P_y}$$

New equilibrium:

Supply: $\frac{P_x}{P_y} = \frac{x}{y}$

Demand: $\frac{P_x}{P_y} = \frac{y}{9x}$

PPF Example: Solution

- e) If consumer preferences switch to favor good y as $U(x, y) = x^{0.1}y^{0.9}$, determine the new demand functions and equilibrium with the old PPF

Thus, $9x^2 = y^2$

Replacing this condition on the old PPF:

$$x^2 + 9x^2 = 100$$

$$x^2 = 10$$

$$x^* = \sqrt{10}, y^* = 3\sqrt{10}, \quad \frac{P_x}{P_y} = \frac{1}{3}$$

Welfare Economics

Competitive Markets

14. Welfare Economics

Relationships described by competitive model are so **complex** that it is hard to believe that any desirable outcome will emerge from **chaos**

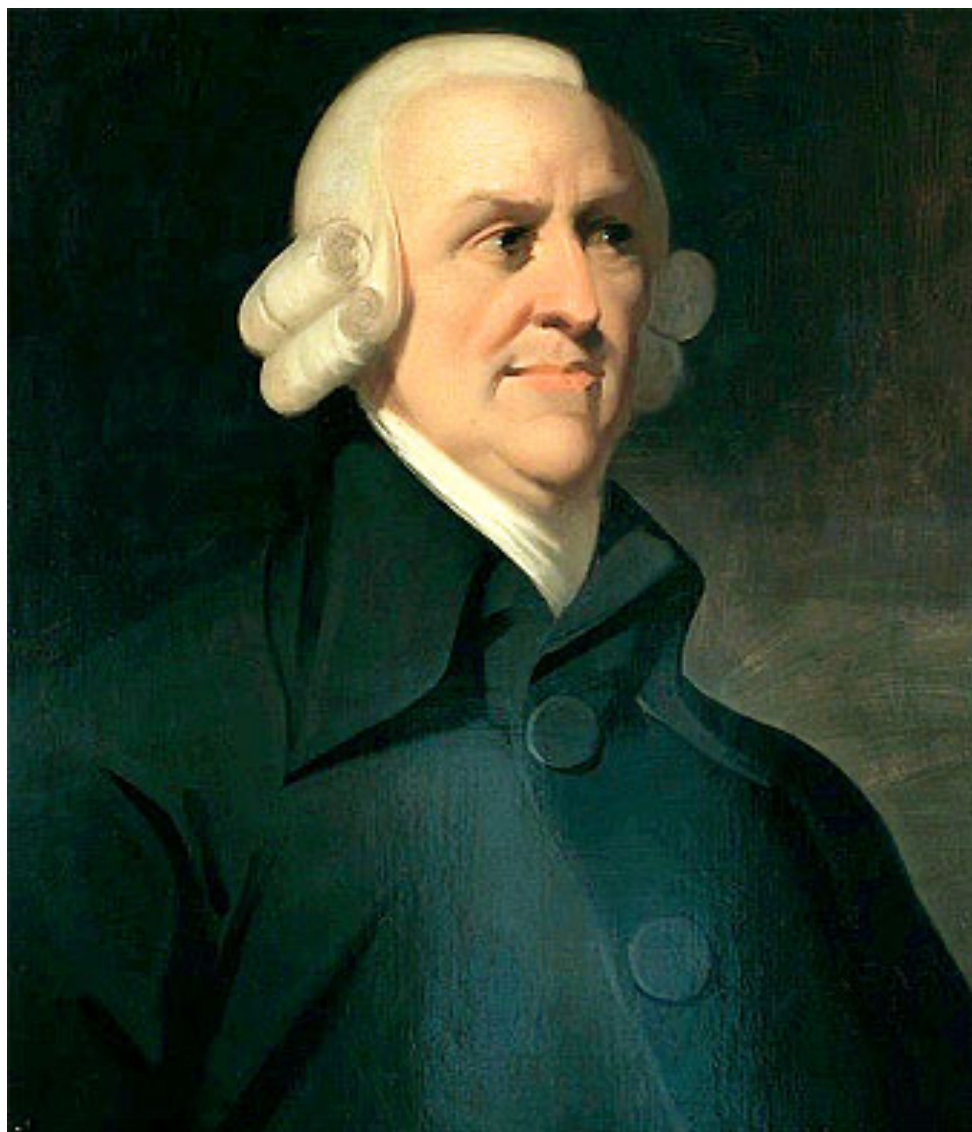
Thus, some thought that market forces were chaotic and that human societies can do better through careful planning

→ **Central Planning**
(predominant in 18th Century)

Who challenged this view?

14. Welfare Economics

Adam Smith
(1723–1790)



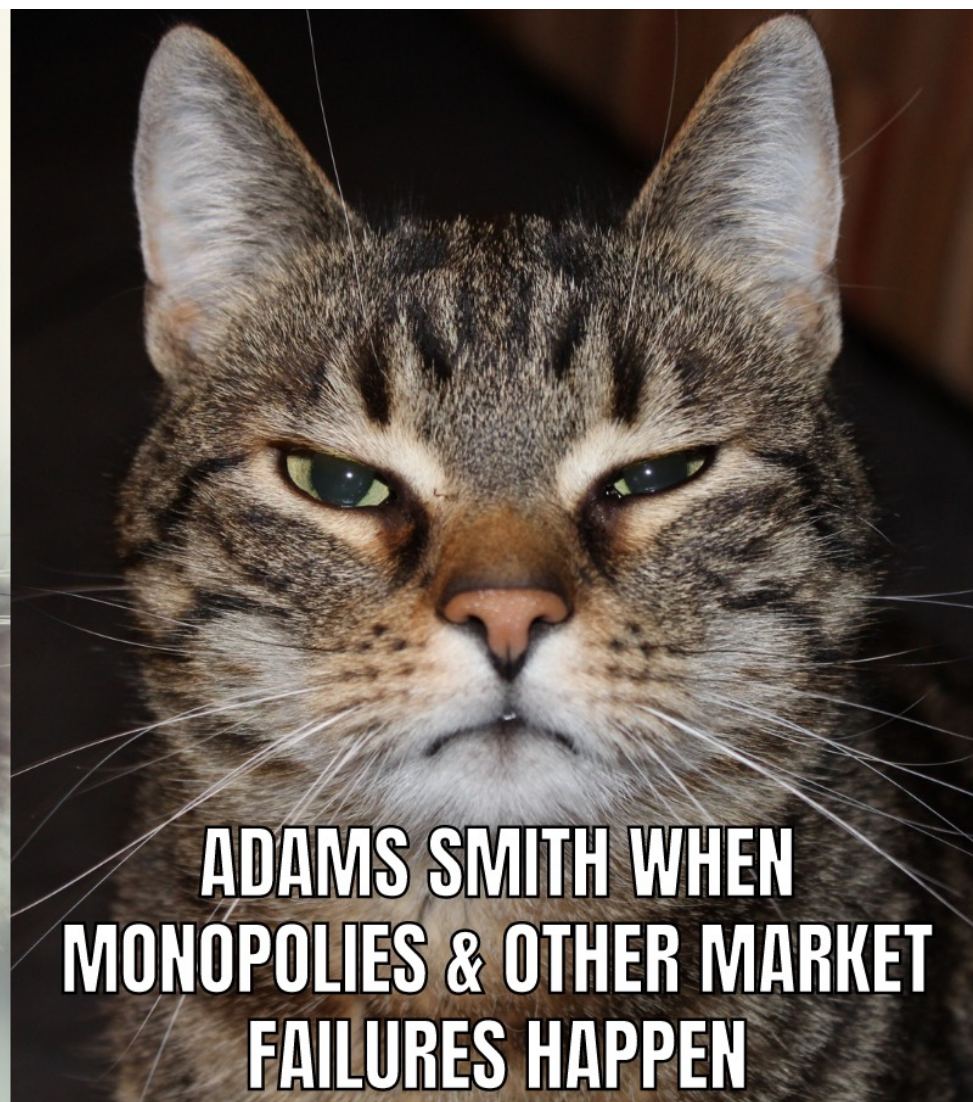
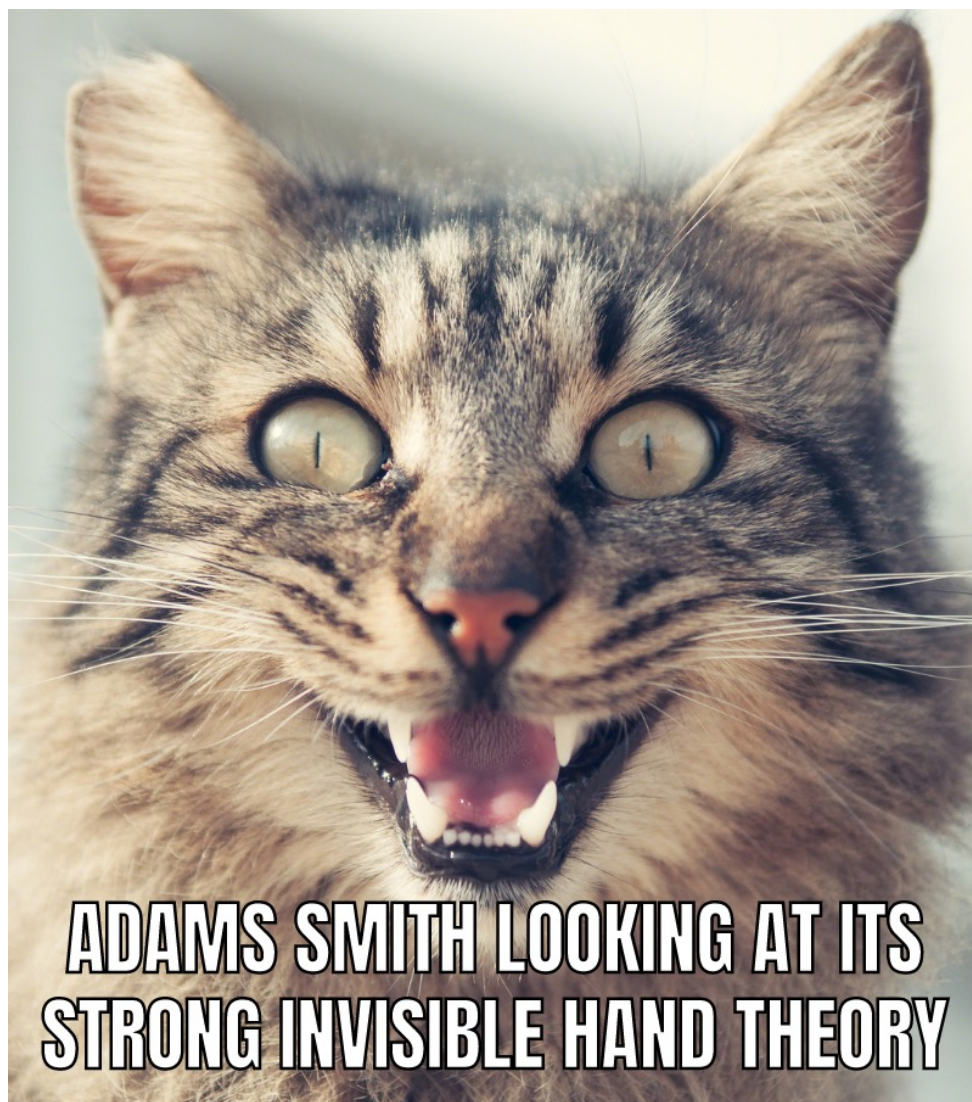
The Muir portrait at the [Scottish National Gallery](https://www.scottishnationalgallery.org.uk/)¹⁹

14. Welfare Economics

Smith's invisible hand hypothesis: invisible hand ensured that **resources would find their way to where they were most valued** → enhancing the wealth of the nation

Smith's invisible hand provided impetus for the **First Theorem of Welfare Economics:**

“There is close correspondence between efficient allocation of resources and the competitive pricing of these resources”



Economic Efficiency

Competitive Markets

15. Economic efficiency

Definition: **Pareto efficient allocation**

An allocation of resources is Pareto efficient if it is not possible (through further reallocations) to make one person better-off without making someone else worse-off

Vilfredo Pareto
(1848-1923)



15. Economic Efficiency

Economic efficiency criteria

In order to be in an efficient allocation, our economy must satisfy four conditions:

1. Efficiency in production

- a. Efficient allocation of resources among firms
- b. Be in the PPF

2. Efficiency in consumption

3. Efficiency in the output mix

15. Economic Efficiency:

Efficiency in production

1. Efficiency in production

In order to achieve technical efficiency, resources must be allocated correctly among firms. Intuitively, resources should be allocated to those firms where they can be most efficiently used.

a) Efficient allocation of resources among firms:

Resources must be allocated to those firms where they can be most efficiently used → Marginal Physical Product of any resource in the production of a particular good is the same, independent of which firm produces that good:

$$\frac{\partial f_1}{\partial k_1} = \frac{\partial f_2}{\partial k_2} \wedge \frac{\partial f_1}{\partial l_1} = \frac{\partial f_2}{\partial l_2}$$

15. Economic Efficiency:

Efficiency in production

1. Efficiency in production

a) Efficient allocation of resources among firms (PROOF)

Suppose there are two firms producing the same good (x) and their production functions are given by $f_1(k_1, l_1)$ and $f_2(k_2, l_2)$

Total supplies of capital and labour are given by $\bar{k}$ and $\bar{l}$. The allocation problem is then to maximize:

$$x = f_1(k_1, l_1) + f_2(k_2, l_2)$$

Subject to the constraints:

$$\begin{aligned} k_1 + k_2 &= \bar{k} \\ l_1 + l_2 &= \bar{l} \end{aligned}$$

15. Economic Efficiency: Efficiency in production

1. Efficiency in production

a) Efficient allocation of resources among firms (PROOF)

Substituting the constraints, the maximization problem becomes

$$x = f_1(k_1, l_1) + f_2(\bar{k} - k_1, \bar{l} - l_1)$$

First order conditions for a maximum are:

$$\frac{\partial x}{\partial k_1} = \frac{\partial f_1}{\partial k_1} + \frac{\partial f_2}{\partial k_1} = \frac{\partial f_1}{\partial k_1} - \frac{\partial f_2}{\partial k_2} = 0 \quad \text{and}$$

$$\frac{\partial x}{\partial l_1} = \frac{\partial f_1}{\partial l_1} + \frac{\partial f_2}{\partial l_1} = \frac{\partial f_1}{\partial l_1} - \frac{\partial f_2}{\partial l_2} = 0$$

$$\frac{\partial f_1}{\partial k_1} = \frac{\partial f_2}{\partial k_2} \wedge \frac{\partial f_1}{\partial l_1} = \frac{\partial f_2}{\partial l_2}$$

Activity 1

Gains from efficiently allocating labour

Examine the quantitative gains in output from allocating resources efficiently:

If there are two rice farms with the following production function:

$$q = k^{1/4}l^{3/4}$$

One rice farm is more mechanized than the other. If capital for the first farm is given by $k_1=16$ and for the second $k_2=625$ and total labour supply is 100:

- Calculate the production function for each farm
- Find the total quantity of rice produced if we impose an equal allocation of labour for these two farms
- Find the efficient allocation of labour for each farm and the total output for this case

15. Economic Efficiency:

Efficiency in production

1. Efficiency in production

Although equality of marginal productivities will ensure the efficient allocation of resources among firms producing any one good, that condition is not enough to ensure that inputs are allocated efficiently among firms producing different goods.

b) Economy is efficient in production if it is on its PPF:

- An allocation of resources is **efficient in production** (or “technically efficient”) if no further reallocation would permit more of one good to be produced without necessarily reducing the output of some other good

$$RTS_{k,l}^x = RTS_{k,l}^y$$

Technical efficiency is a precondition (we must be in the PPF)
BUT it does not guarantee Pareto efficiency.

15. Economic Efficiency:

Efficiency in production

Let's demonstrate mathematically that an economy is efficient in production if it is on its PPF:

Two goods are being produced (x and y). The production function of x is given by $x = f(k_x, l_x)$ and for good y by $y = g(k_y, l_y)$.

Total availability of the inputs is constrained by $\bar{k} = k_x + k_y$ and $\bar{l} = l_x + l_y$.

15. Economic Efficiency:

Efficiency in production

Achieving technical efficiency is to maximize the value of x for any specified value of y , say $\bar{y}$:

$$\mathcal{L} = f(k_x, l_x) + \lambda [g(\bar{k} - k_x, \bar{l} - l_x) - \bar{y}]$$

First order conditions are:

$$\left. \begin{aligned} \frac{\partial \mathcal{L}}{\partial k_x} &= f_{k_x} - \lambda g_{k_x} = 0 \\ \frac{\partial \mathcal{L}}{\partial l_x} &= f_{l_x} - \lambda g_{l_x} = 0 \end{aligned} \right\} \begin{aligned} \frac{f_k}{f_l} &= \frac{g_k}{g_l} \\ RTS_{k,l}^x &= RTS_{k,l}^y \end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = g(\bar{k} - k_x, \bar{l} - l_x) - \bar{y} = 0$$

15. Economic Efficiency:

Efficiency in consumption

If we are talking about economic efficiency, we should also take into account consumer's preferences.

Mathematical proof:

Two individuals economy, each with a different utility (U^1 and U^2)

Two goods (x and y) are produced in a fixed amount, where x_i is the quantity of x consumed by individual i

15. Economic Efficiency: Efficiency in consumption

We want to solve the following problem:

$$\text{Max}_{\{x_1, x_2, y_1, y_2\}} U^1(X_1, Y_1)$$

s.to:

$$X_1 + X_2 = \bar{X}$$

$$Y_1 + Y_2 = \bar{Y}$$

$$U^2(X_2, Y_2) = \bar{U}$$

$$\Leftrightarrow \text{Max}_{\{X_1, Y_1\}} L = U^1(X_1, Y_1) + \lambda(\bar{U} - U^2(\bar{X} - X_1, \bar{Y} - Y_1))$$

15. Economic Efficiency: Efficiency in consumption

$$\Leftrightarrow \text{Max}_{\{X_1, Y_1\}} L = U^1(X_1, Y_1) + \lambda(\bar{U} - U^2(\bar{X} - X_1, \bar{Y} - Y_1))$$

First order conditions:

$$\frac{\partial L}{\partial X_1} = 0 \Leftrightarrow \frac{\partial U^1}{\partial X_1} - \lambda \frac{\partial U^2}{\partial X_2} \frac{\partial X_2}{\partial X_1} = 0 \Leftrightarrow \frac{\partial U^1}{\partial X_1} + \lambda \frac{\partial U^2}{\partial X_2} = 0 \Leftrightarrow \lambda = -\frac{MU_X^1}{MU_X^2}$$

$$\frac{\partial L}{\partial Y_1} = 0 \Leftrightarrow \frac{\partial U^1}{\partial Y_1} - \lambda \frac{\partial U^2}{\partial Y_2} \frac{\partial Y_2}{\partial Y_1} = 0 \Leftrightarrow \frac{\partial U^1}{\partial Y_1} + \lambda \frac{\partial U^2}{\partial Y_2} = 0 \Leftrightarrow \lambda = -\frac{MU_Y^1}{MU_Y^2}$$

Hence:

$$MRS_{X,Y}^1 = \frac{MU_X^1}{MU_Y^1} = \frac{MU_X^2}{MU_Y^2} = MRS_{X,Y}^2$$

15. Economic Efficiency: Efficiency in the Output Mix

But one thing is **technical efficiency**, and another is **Pareto optimality** → Demand must also be brought into the story → to tie together individual preferences and production possibilities.

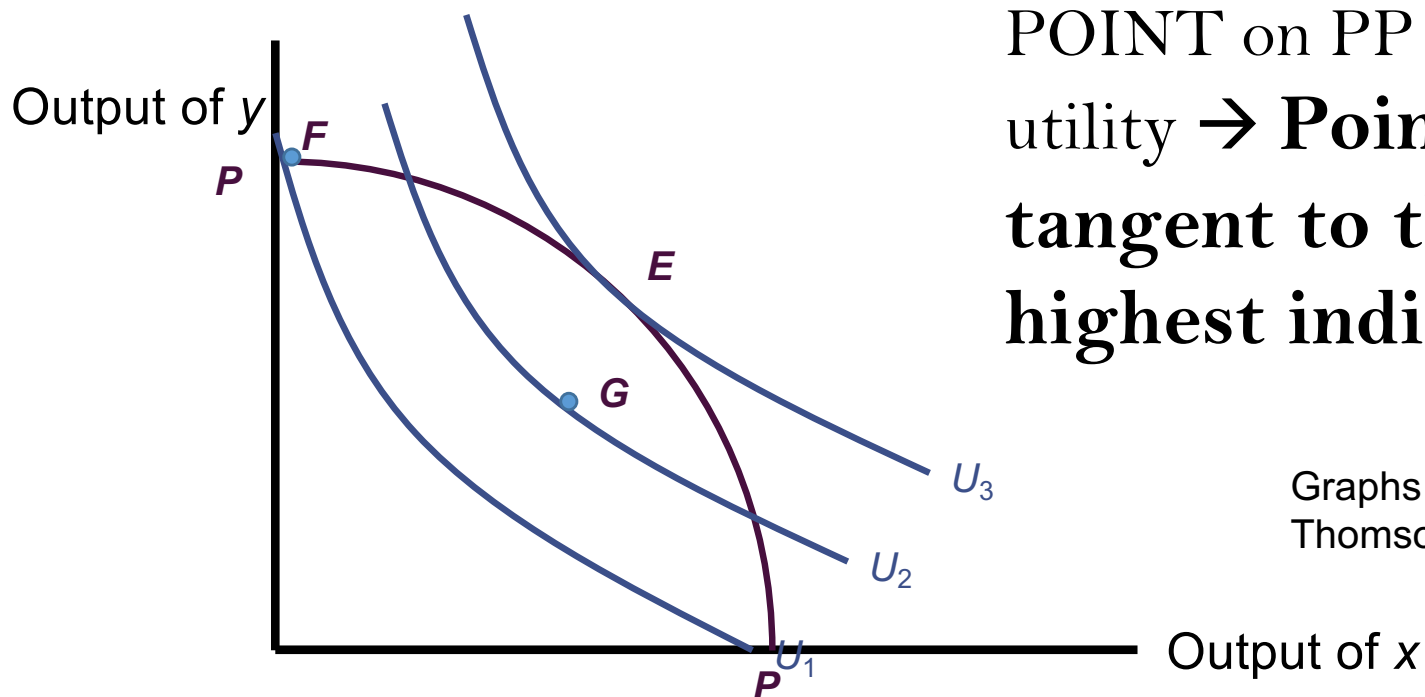
We need to ensure that the suitable goods produced → MRS for any two goods must be equal to the RPT of the two goods.

$$\text{MRS}_{x,y} = \text{RPT}_{x,y}$$

15. Economic Efficiency: Efficiency in the Output Mix

A one-person economy (Robison Crusoe) produces only two goods (x, y). Any point on PP represents a point of technical efficiency (is PPF)

By superimposing an individual's indifference map only **ONE POINT** on PP provides maximum utility → **Point E** where **PP** is tangent to the individual's highest indifference curve



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Activity 2

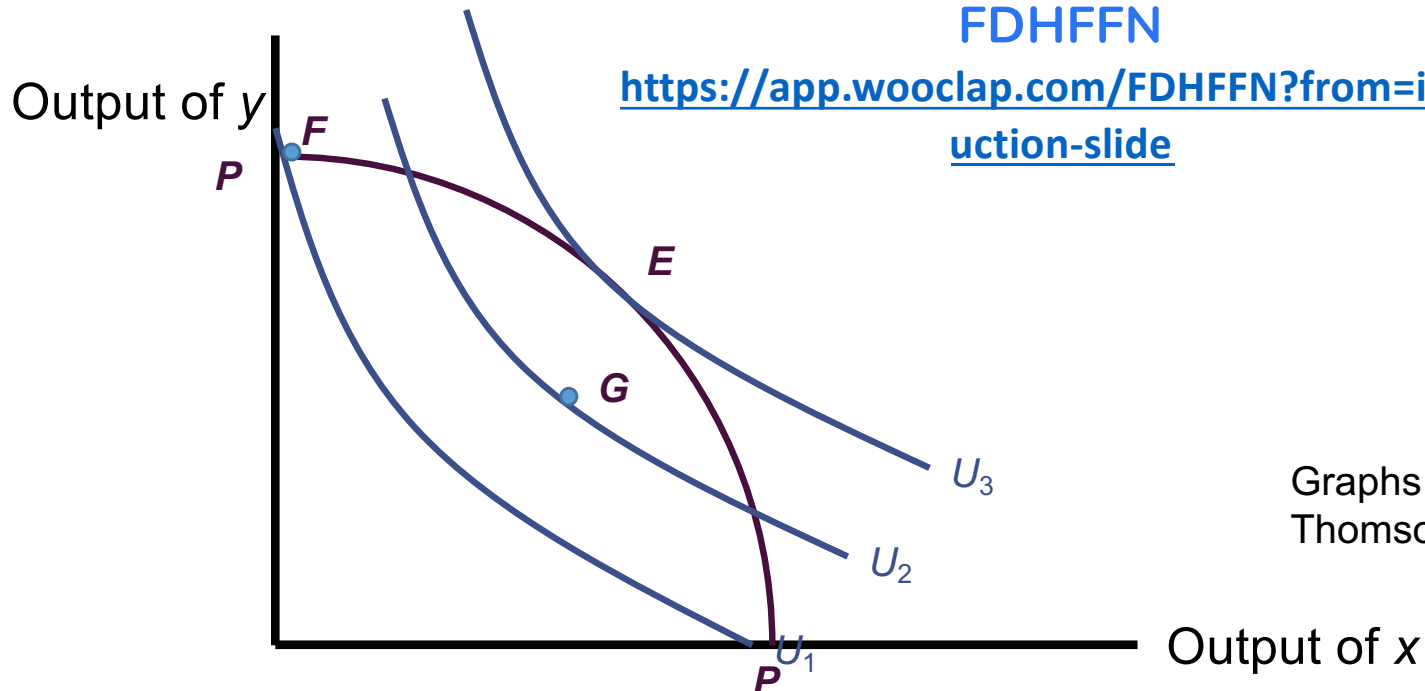
We now know that point E is the efficient one in this example. Imagine that, for some reason, we cannot produce E.

Which point should be preferred, F or G?

Código de evento

FDHFFN

<https://app.wooclap.com/FDHFFN?from=instruction-slide>



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15. Economic Efficiency: Efficiency in the Output Mix

Mathematical proof

One person economy (Robison Crusoe), produces only two goods (x,y) and utility function is $U(x,y)$. Let's suppose that this society's PPF can be written in implicit form as $T(x,y) = 0$

Robinson's problem is to maximize utility subject to his production constraint (PPF):

$$\mathcal{L} = U(x, y) + \lambda[T(x, y)]$$

First order conditions for an interior maximum are:

$$\left. \begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= \frac{\partial U(x, y)}{\partial x} + \lambda \frac{\partial [T(x, y)]}{\partial x} = 0, \\ \frac{\partial \mathcal{L}}{\partial y} &= \frac{\partial U(x, y)}{\partial y} + \lambda \frac{\partial [T(x, y)]}{\partial y} = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= T(x, y) = 0 \end{aligned} \right\} \begin{aligned} \frac{\partial U / \partial x}{\partial U / \partial y} &= \frac{\partial T / \partial x}{\partial T / \partial y} \\ \Leftrightarrow MRS_{x,y} &= -\frac{dy}{dx}(\text{along } T) = RPT_{x,y} \end{aligned}$$

15. Economic Efficiency

(summary)

Recap: We are looking for **economic efficiency**, and it needs:

1. Efficiency in production (technical efficiency)

a) Be in PPF $\rightarrow \frac{\partial f_1}{\partial k_1} = \frac{\partial f_2}{\partial k_2}, \frac{\partial f_1}{\partial l_1} = \frac{\partial f_2}{\partial l_2}$

- b) Efficient allocation of resources among firms $\rightarrow$

$$\frac{f_k}{f_l} = \frac{g_k}{g_l} \Leftrightarrow RTS_{l,k}^x = RTS_{l,k}^y$$

2. Efficiency in consumption

$$MRS_{x,y}^1 = MRS_{x,y}^2$$

3. Efficiency in output mix

$$MRS_{x,y} = -\frac{dy}{dx}(\text{along } T) = RPT_{x,y}$$

First Theorem of Welfare Economics

Competitive Markets

16. First Theorem of Welfare Economics

Attaining a Pareto efficient allocation of resources requires that the **rate of the trade-off between any two goods should be the same** for all economic agents

→ In a perfectly competitive economy **ratio of prices** provides this expected rate of trade-off to which agents will adjust

- Prices are treated as fixed parameters both in an individual's utility-maximizing and in a firm's profit-maximizing
- All agents face the same prices, all trade-off rates will be equalized, and an efficient allocation will be achieved

16. First Theorem of Welfare Economics

All this is the First Theorem of Welfare Economics

- There is **efficiency in production** and in the **output mix**
- “Invisible hand” ensures not only a technically efficient production but also that the forces of supply and demand lead to the Pareto efficient output combination, given a price ratio of P_x^*/P_y^*

Second Theorem of Welfare Economics

Competitive Markets

17. Distribution and the Second Theorem of Welfare Economics

First Theorem ensures that competitive markets will achieve efficient allocations, but there is **no guarantee that these allocations will achieve any sort of fair distribution of welfare among individuals.**

Amartya Sen said that an allocation of resources might be Pareto efficient “even when some people are rolling in luxury and others are near starvation, as long as the starvers cannot be *better off without cutting the pleasures of the rich....* In short, a society can be Pareto optimal and still be *perfectly disgusting.*”



17. Distribution and the Second Theorem of Welfare Economics

So, what might happen if initial endowments are skewed in favour of some economic actor?

Second Theorem of Welfare

- Any desired distribution of welfare among individuals in an economy can be achieved efficiently through competitive pricing IF initial endowments are adjusted properly.
- Efficiency issues $\neq$ equity issues.
- Economists frequently argue for using the efficiency property of competitive prices to “make the pie as big as possible” and then adjusting the resulting distribution to be “fair” through the use of lump-sum transfers (such as taxes, and subsidies)



Equity

**Justice in the distribution
of well-being among the
members of society.**

**The property of
distributing economic
prosperity evenly among
members of a society**

EQUALITY VERSUS EQUITY



In the first image, it is assumed that everyone will benefit from the same supports. They are being treated equally.



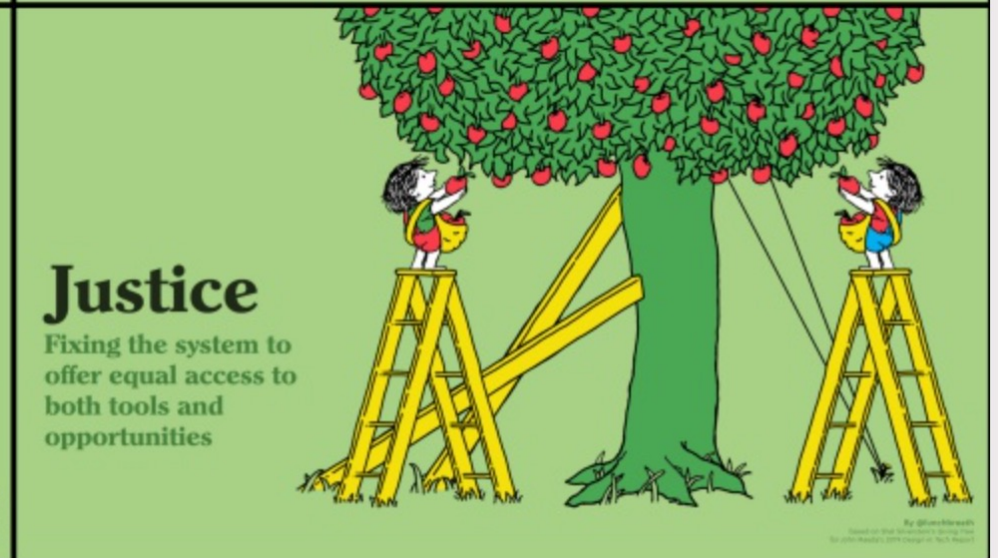
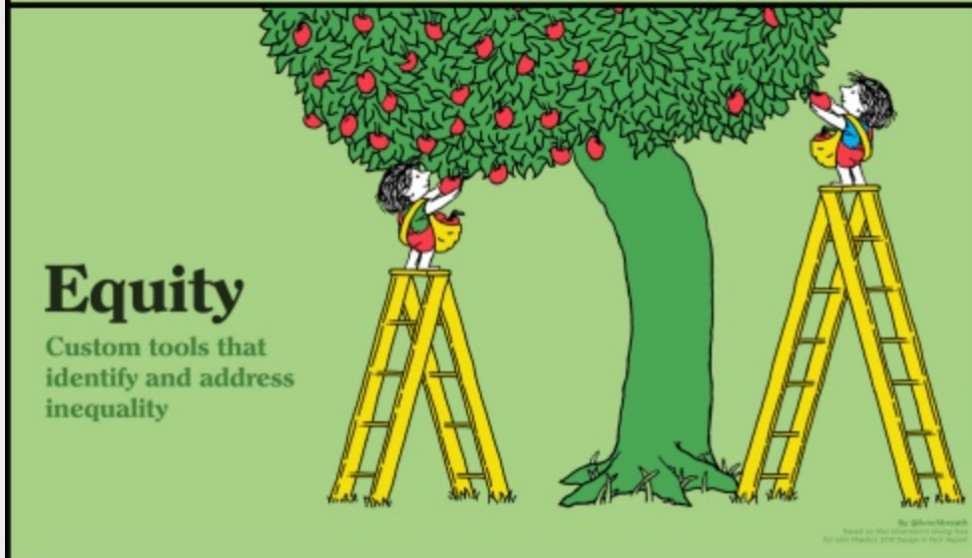
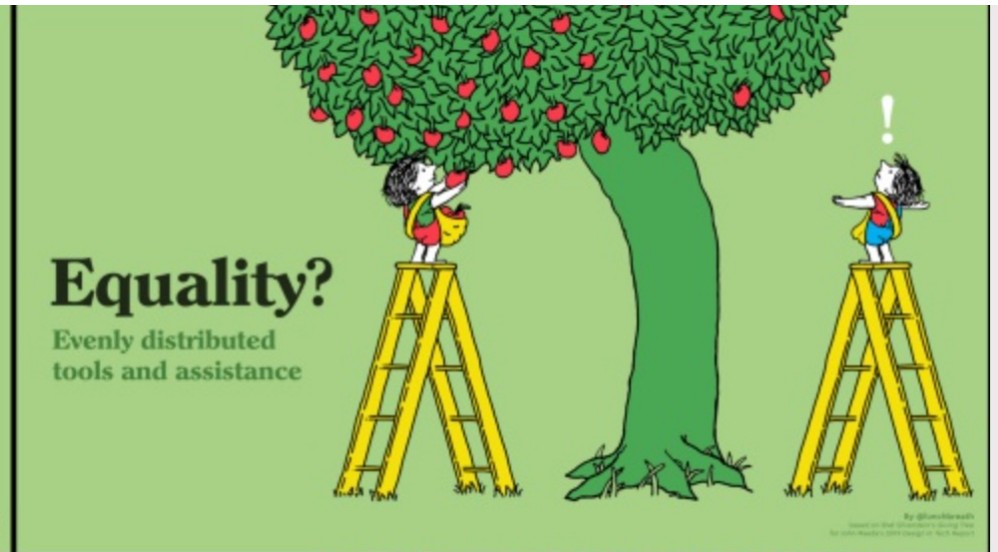
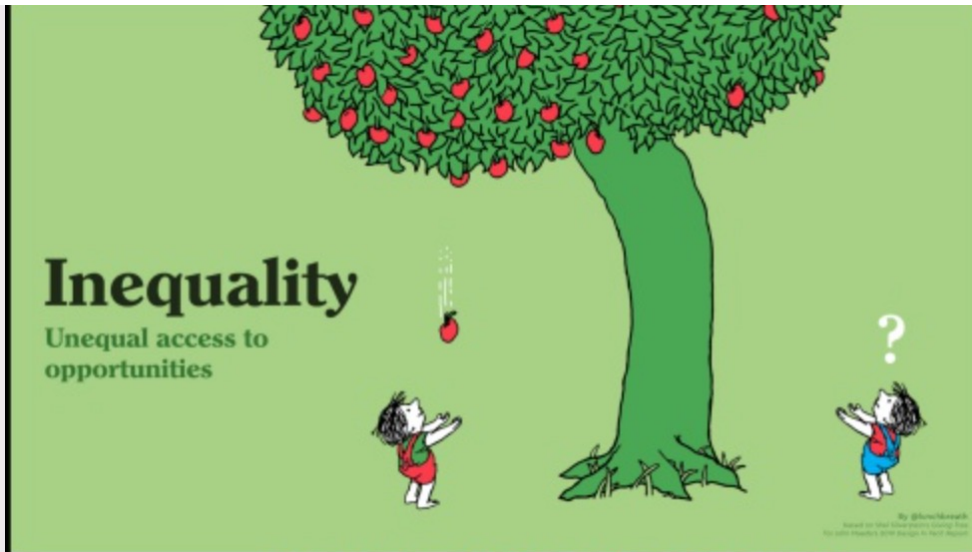
In the second image, individuals are given different supports to make it possible for them to have equal access to the game. They are being treated equitably.



In the third image, all three can see the game without any supports or accommodations because the cause of the inequity was addressed. The systemic barrier has been removed.

En esta versión, la valla se sustituye por una alambrada y todo el mundo puede ver el partido, es decir, se busca una solución a la causa de la inequidad

(2012)



Cuatro ilustraciones sobre desigualdad, igualdad, equidad y justicia. Tony Ruth

Individual Activity

On a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it in your own words.

If you have any **questions** about what we discussed today, write them down too. We Will go through them at the beginning of the next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.